UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	A	Which data type stores a whole number?	int stores whole numbers; float stores real values.
A2	D	What is displayed by the following code?	The loop displays 2, 4, 6 and 8 without spaces.
A3	B	Which item is NOT a programming paradigm?	Debugging is an activity, not a programming paradigm.
A4	C	Given the declaration, what is displayed?	word[3] is 't' because array indexing starts at zero.
A5	A	Which of the following are stream operators?	<< inserts into a stream and >> extracts from one.
A6	D	Which structure chooses between two alternative paths?	if...else performs two-way selection.
A7	D	Which tool represents an algorithm using standard graphical symbols?	A flowchart uses standard symbols such as terminals, processes and decisions.
A8	A	What does the % operator return?	% is the modulus operator and returns the remainder.
A9	B	Which of the following declarations are valid?	a and c are valid array declarations.
A10	C	Which statement is most suitable for selecting among several constant cases?	switch selects a case based on a constant expression.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define algorithm and pseudocode.

An algorithm is a finite sequence of precise steps that solves a problem. Pseudocode is a structured English-like description of an algorithm.

Marking: Award 1 mark for each correct definition.

(3 marks)

b) Explain THREE types of programming errors and give one example of each.

Syntax error: missing semicolon. Runtime error: division by zero. Logic error: using < instead of <= in a condition.

Marking: Award 0.5 mark for each type and 0.5 mark for a valid example.

(2 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart for the ticket price selection.

Start -> read age -> if age < 0 display invalid; else if age <= 12 price = 15; else if age <= 17 price = 25; else price = 40 -> display price -> end.

Marking: Award marks for validation, all ranges, output and flow.

(5 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

Use if/else if with the three ranges and an invalid-age branch.

Marking: Award marks for validation, range conditions, assignment and output.

(8 marks)

QUESTION 3

Topic: data types

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

int age; double price; const double DISCOUNT = 0.10; char grade;

Marking: Award marks for correct types, names and const initialisation.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

cin >> age >> price; price = price - price * DISCOUNT; cout << price;

Marking: Award marks for input, discount expression and output.

(5 marks)

QUESTION 4

Topic: input output

a) Trace the program and show the exact output.

1\n12\n123

Marking: Award marks for each line and the exact row lengths.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

The inner loop must initialise j = 1 for every row and the outer loop must run to 4. Correct code: for (int i=1;i<=4;i++){for(int j=1;j<=i;j++) cout<<j; cout<<endl;}

Marking: Award marks for each error and the corrected loop.

(5 marks)

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QUESTION 5

Topic: arrays

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

string sports[5] = {"Swimming","Badminton","Athletics","Cycling","Hockey"}; int medals[5] = {8,12,7,5,9};

Marking: Award marks for both declarations, correct sizes and related indices.

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

Loop i=0..4 to display sports[i] and medals[i], accumulate total, average = double(total)/5.

Marking: Award marks for paired display, total, average and output.

(10 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    int code, quantity;
    double price = 0, subtotal, discount = 0;
    string name;
    cout << "Item code (1 coffee, 2 tea, 3 juice): ";
    cin >> code;
    cout << "Quantity: ";
    cin >> quantity;
    if (code == 1) { name = "Coffee"; price = 8.50; }
    else if (code == 2) { name = "Tea"; price = 6.00; }
    else if (code == 3) { name = "Juice"; price = 9.50; }
    else { cout << "Invalid item code" << endl; return 1; }
    subtotal = price * quantity;
    if (subtotal >= 50) discount = subtotal * 0.10;
    cout << fixed << setprecision(2);
    cout << "Item: " << name << endl;
    cout << "Quantity: " << quantity << endl;
    cout << "Subtotal: RM" << subtotal << endl;
    cout << "Discount: RM" << discount << endl;
    cout << "Amount to pay: RM" << subtotal - discount << endl;
    return 0;
}
```

Marking: Award 3 marks for input, 5 for price/subtotal, 4 for discount, 2 for formatting and 1 for completeness.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
```

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```
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

END OF MARKING SCHEME

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